Knowledge organiser · a fair rover test

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

I can carry out a comparative fair test, use repeated distance results for a conclusion and improve the method.

We test one rover on two surfaces. We keep each run. We use the results to say what we found.

Change the surface. Keep the rover, start, ramp and release the same. Release with no push. Measure from the same line each time.

Repeat means do the same test again. Keep each valid result, even if it is a surprise. A failed method needs a note and a repair.

Surface	Run 1 (cm)	Run 2 (cm)	Run 3 (cm)
Smooth	48	51	49
Fabric	27	29	28

These are fixed teaching data. Each smooth run went farther. Friction opposed the motion less on the smooth surface in this model.

A conclusion uses the pattern. Improve one method step for the next test. A classroom model cannot show all real rover conditions.

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Word page

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Word	Plain bridge	Meaning
repeat	same method again	another measurement made under the same condition
conclusion	what the results support	an answer to the question that uses the evidence
limitation	what the test cannot show	a feature that restricts how far the result can be applied

A repeat uses the same method. A conclusion says what the results show. A limit says what this test cannot show.

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Arrival · Supported

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

Recall the surface plan. Change one factor, measure distance and keep the other key parts the same.

1. Point to the one variable changed in the surface test.
2. Point to centimetres as the distance unit.
3. Count the three result boxes for each condition.
4. Complete: Keep every result, even if it is ___.

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Arrival · Independent

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

Recall the surface plan. Change one factor, measure distance and keep the other key parts the same.

1. State the surface-test independent and dependent variables.
2. Name two controls from the plan.
3. Explain why the rover is released without a push.
4. Predict which surface gives the greater distance and why.

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Arrival · Aspire

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

Recall the surface plan. Change one factor, measure distance and keep the other key parts the same.

1. Explain what to do with an unexpected but valid result.
2. Calculate a range and explain how it shows variation.
3. Name one method repair if a run leaves the lane.
4. State one limit of applying the result to a planetary rover.

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Starter · keep each result

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Smooth-surface results are 48, 51 and 49 cm. What should happen to the 51 cm result?

A. Keep it with the other valid repeats

B. Delete it because it is not identical

C. Keep only 51 because it is the furthest

Keep your first idea. Use the model to check it later.

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PREDICT · commit before testing

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

I think the rover will go farther on smooth, because…

Keep this first guess. Check it with the test.

Make a guess before the test. Say which surface you think will let the rover go farther. Give a reason. Keep the guess so you can check it with the six runs.

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TEST · repeat the same method

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.



Fixed teaching data. Keep each of the six values.

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RECORD · pattern, conclusion, improvement

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.



Each fabric run is shorter in this set. Use all six runs to check the claim. Friction opposed the motion more on fabric. Say one way to keep the next test fair.

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I do · method and safe repeats

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Check the clear lane and low ramp. Use the same rover and marked start. Predict what will happen before you let go.

Release without a push, measure from the same line and record immediately. Repeat three times before changing condition.

Keep every valid result. If the method fails, preserve the first note, repair the setup and repeat the run.

Teacher checks the ramp and clear lane. Release without pushing; keep hands, feet and faces away from the moving rover.

One person releases. One watches and records. Swap roles. Both learners keep their own results and reason.

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Results desk · keep and compare

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Surface	Run 1 (cm)	Run 2 (cm)	Run 3 (cm)
Smooth	48	51	49
Fabric	27	29	28

Fixed teaching data, not your own new test. Keep 51 cm with the other valid smooth results.

Smooth mean: (48 + 51 + 49) ÷ 3 = 49.33… cm. About 49 cm. Fabric mean: (27 + 29 + 28) ÷ 3 = 28 cm.

Smooth range: 51 − 48 = 3 cm. Fabric range: 29 − 27 = 2 cm. Each smooth value is greater than each fabric value.

Use a release gate to keep the start the same. Test it with the same rover. These runs show one classroom test. They cannot show all real rover conditions.

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We do · choose the record

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Which recording choice gives the strongest evidence?

A. Keep all three valid runs for each surface. Use the same rule to measure each run.

B. Keep only the furthest result from each condition.

C. Change a result when it does not match the prediction.

Show a choice and give a reason. Both learners get a turn.

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I do 2 · pattern, reason and limit

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Fixed teaching data: smooth 48, 51, 49 cm. Fabric 27, 29, 28 cm. Each smooth run went farther.

The mean is about 49 cm on smooth and 28 cm on fabric. Smooth runs span 48 to 51 cm. Fabric runs span 27 to 29 cm. These spans do not overlap.

Range means highest minus lowest. Smooth range is 51 − 48 = 3 cm. Fabric range is 29 − 27 = 2 cm.

The rover went farther on smooth. Lower friction opposed its motion less. This is the conclusion for this model.

Use the fixed six results if you cannot run the test. Say that they are fixed teaching data. Keep each value.

We tested one rover on two surfaces. A real rover may face other ground and gravity. Its design may differ too. Our test cannot show each of these effects. This is a model limit.

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Rover results lab · smooth

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Sort each card into Smooth, Fabric or Conclusion. Keep the pattern. Then choose a way to improve the next test.

48, 51, 49 cm

These are the three fixed smooth runs. Keep all three.

Range: 3 cm (48–51 cm interval)

Highest minus lowest: 51 − 48 = 3 cm. This is the smooth range.

Keep the pattern and your conclusion. Name one change to make the next test better. Say what this model cannot show.

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Rover results lab · fabric

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Sort each card into Smooth, Fabric or Conclusion. Keep the pattern. Then choose a way to improve the next test.

27, 29, 28 cm

These are the three fixed fabric runs. Keep all three.

Range: 2 cm (27–29 cm interval)

Highest minus lowest: 29 − 27 = 2 cm. This is the fabric range.

Keep the pattern and your conclusion. Name one change to make the next test better. Say what this model cannot show.

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Rover results lab · conclusion

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Sort each card into Smooth, Fabric or Conclusion. Keep the pattern. Then choose a way to improve the next test.

The rover went farther on smooth.

Friction opposed the motion less on smooth. Use all six runs to check this claim.

We tested one rover on two surfaces.

A real rover may have a different design. Ground and gravity may differ too. Our test cannot show all of those effects.

Keep the pattern and your conclusion. Name one change to make the next test better. Say what this model cannot show.

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Science glitch · keep valid repeats

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Keep the furthest result and discard the rest because it shows the rover's best performance.

A. Keep the claim: only the best result matters.

B. Repair it: keep every valid repeat so the pattern and variation remain visible.

C. Change it: write one average and erase all raw results.

This claim is a challenge to check. Give the choice that keeps the full evidence.

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Independent · Supported

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

Carry out or direct the fair rover test, record two conditions with three repeats, then give a conclusion, improvement and limitation.

Direct each run, match the six values to conditions and complete an oral or symbol-supported conclusion.

Adult handles equipment; use pre-labelled rows and the frame: On ___ it went ___ because ___.

Teacher checks the ramp and clear lane. Release without pushing; keep hands, feet and faces away from the moving rover.

Use a safe real test, direct an adult, or use the fixed table. Mark which route you used. Keep the same science decisions.

Choose one route within the full 15 minutes. Keep each raw result.

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Independent · Independent

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

Carry out or direct the fair rover test, record two conditions with three repeats, then give a conclusion, improvement and limitation.

Complete three safe repeats per condition, record all results and write a conclusion plus one improvement.

Check the method against change, measure, keep and repeat before concluding.

Teacher checks the ramp and clear lane. Release without pushing; keep hands, feet and faces away from the moving rover.

Use a safe real test, direct an adult, or use the fixed table. Mark which route you used. Keep the same science decisions.

Choose one route within the full 15 minutes. Keep each raw result.

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Independent · Aspire

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

Carry out or direct the fair rover test, record two conditions with three repeats, then give a conclusion, improvement and limitation.

Find each mean and range. Say how the runs vary. How far can this model tell us about a real rover?

Give one change to improve the next test. Say why it helps. Keep the raw results as they are.

Teacher checks the ramp and clear lane. Release without pushing; keep hands, feet and faces away from the moving rover.

Use a safe real test, direct an adult, or use the fixed table. Mark which route you used. Keep the same science decisions.

Choose one route within the full 15 minutes. Keep each raw result.

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My test · prediction and all six results

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

My route: real test / direct an adult / fixed teaching data.

I predict _____ because _____.

Change _____. Measure _____. Keep _____ the same.

Surface	Run	Distance (cm)
Smooth	1	_____
Smooth	2	_____
Smooth	3	_____
Fabric	1	_____
Fabric	2	_____
Fabric	3	_____

If the method fails, keep the note. Repair it, then repeat the run. Note any extra run: _____.

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My test · pattern, conclusion and next method

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Pattern in all six results: _____

My conclusion and science reason: _____

One method improvement and why: _____

One model limit and why: _____

What the adult heard: _____

My next response, or why I kept it: _____

Keep the first record. A mean or range goes beside the raw results.

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Exit · Supported

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Which condition gave the greater travel distance?

Use your own result pattern, or name the fixed teaching data. The adult hears your response and names back the science.

What I said: _____

What changed or stayed, and why: _____

Next I will: _____

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

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Exit · Independent

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Use one repeated result pattern to explain the effect of surface on the rover.

Use your own result pattern, or name the fixed teaching data. The adult hears your response and names back the science.

What I said: _____

What changed or stayed, and why: _____

Next I will: _____

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

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Exit · Aspire

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Give one method improvement and one model limitation, explaining why each matters.

Use your own result pattern, or name the fixed teaching data. The adult hears your response and names back the science.

What I said: _____

What changed or stayed, and why: _____

Next I will: _____

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

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